Operational Research: Tutorial 3

Linear Programming 2

The more complicated questions on the tutorial are marked by:

* Medium

** Hard

Pyomo

Notebook (jckantor.github.io)

Exercises

1) **Solve the LP using the Problem-Based approach in Python using the Pyomo library**

maximise $12x + 18y$

subject to:

$$2x + 4y \leq 55$$
$$2x + y \leq 25$$
$$x, y \geq 0$$

   a. **Let x and y be restricted to integer values, solve the Integer Linear Program**
   b. **Let x be a real value and y restricted to an integer, solve the Mixed-Integer Linear Program (MILP)***

Note: You can graphically plot this to validate your solution.

2) *C&S* (Management Consultants) have been employed to help the *Icki-Toffee* Company to develop a popular, economical but more health-conscious selection of candies to add to their scrumptious range of toffees. There has been much publicity that the Government are so worried about obesity that they plan to introduce a coding system for all food products which will inform the public about products containing foods which may be adding to the problem. There are rumours that the Government aim to strongly promote the concept of fruit as a healthy alternative to sweets, so *Icki-Toffee* are very keen to introduce fruit, in the form of fillings, soft centres etc into their range. They are determined to ensure that at least one of their selections of candies does not fall into the worst category.

   The selection chosen must weigh at least 320g and satisfy certain basic restrictions on fat and calorific content. These are outlined below:

   - Total fat content by weight must be at most 45% of the total weight of the selection.
   - Calorific value of the selection must be at most 2850 calories.

   The quantities of fat and calories in each candy are given in Table 1.

   The costs of each candy are given in Table 2.

   Table 2 also lists the results of an in-store survey which Icki-Toffee carried out in several of its retail outlets. The popularity number is measured on the scale where 10 is very popular and 0 not popular.

a) **Formulate a model (using summation notation) given that the aim is to minimise the cost of the selection** *

b) **Obtain the optimal selection via Pyomo** **

c) **Obtain the optimal selection given that the aim is to maximise the popularity of the selection via Pyomo, what is the weight of the selection?** **

d) **Comment on whether these are good selections. Would you be happy?**

Have a think about this yourself, this isn't for me to say, but I will comment on it in the next lecture. Is there a flaw in our modelling, what about our objective, is it really maximising popularity?

Candy - x_i	Fat (g) - F_i	Calories – K_i	Weight (g) - W_i
Buttery butterscotch	20	100	24
Caramel softie	16	100	22
Orange dream	10	80	15
Coconut Ice	7	50	12
Lemon sorbet	6	60	10
Lime sherbet	5	40	10
Strawberry surprise	13	80	14
Raspberry mousse	15	90	17
Ginger toffee	18	80	20
Rose-water posy	-	145	8
Walnut whirl	19	120	20
Grape bunch	-	120	15

Table 1: Table to show content information about each candy

Candy - x_i	Cost - C_i	Popularity - P_i
Buttery butterscotch	12	10
Caramel softie	12	10
Orange dream	12	8
Coconut Ice	15	9
Lemon sorbet	13	5
Lime sherbet	13	6
Strawberry surprise	16	7
Raspberry mousse	16	8
Ginger toffee	15	10
Rose-water posy	10	3
Walnut whirl	18	8
Grape bunch	20	4

Table 2: Cost and popularity data.

Sensitivity: Internal